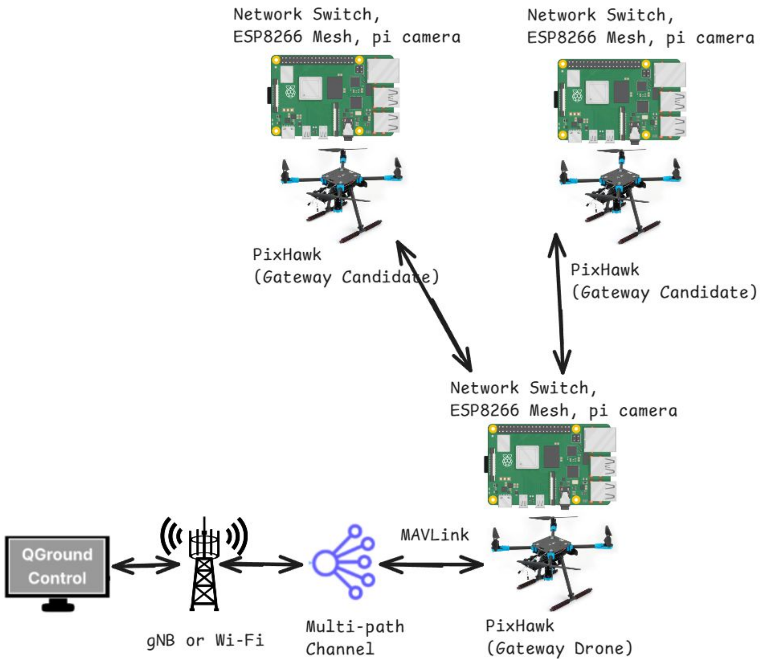
Drone Garage

4/21/26

Forwarding Follower Camera Data to Leader

- esp_rec.py, bs_rec.py
  - Esp_rec.py allows for a pi, whose esp is the leader, to take in data from all other follower nodes in the mesh (including itself); particularly designed for camera frame information; further, it forwards its data to the basestation
  - Bs_rec.py reads this information and prints to the screen, highlighting accuracy in messages
  - Tests on following slide

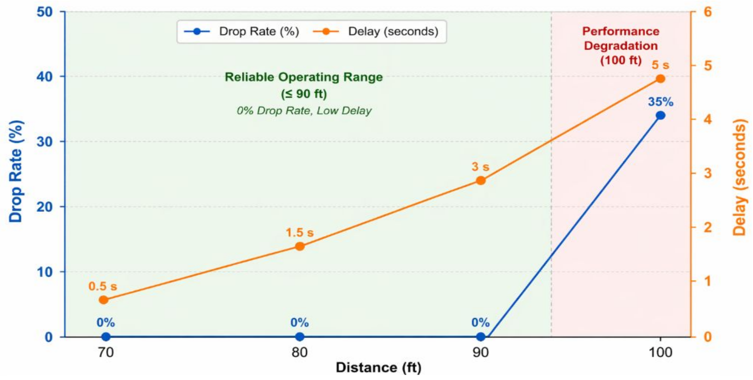


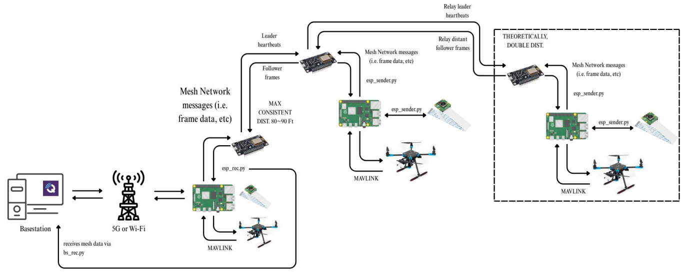
Testing w/ Obstacles and LoS Blockage - 2 Node

- Firstly reconfiguring pi camera to not be so zoomed in
  - Set sensor size in esp_sender accordingly
    - For the picamera v2 configuration settings → **sensor={“output_size”: (3280, 2464)}**
- As of writing, start up esp_rec.py BEFORE powering the leader pi (whichever has lowest meshID), then power up esp_sender on the other pi and stuff
- Results of the tests can be found in bs_detect1.txt – **JUST FOLLOWER DATA NO LEADER**
  - Tried several visual obfuscating methods
  - Obstacles
    - The model identifies objects with high confidence (75~80%; dipped into ~50% like once or twice)
    - The model struggles greatly in identifying objects that are even **slightly** covered
      - But when it does identify things it is accurate in it
    - The model also struggles in identifying multiple objects in a frame
      - Not that it can’t but in cases where there are multiple identifiable objects most of the time it sees only one object and like only a handful of frames it detected two
      - Frame issue? FoV issue?
  - Movement
    - The model can identify objects moving in and out of frame albeit for only a handful of frames
      - Optimization needs to be done on frame rate
        - Modify configuration settings? Redo calibration?

Testing Range - 2 Node

- Max Operable Range: 80~90 Ft (Whatever the distance is from the window of the lab to a couple feet from the sidewalk)
  - Packets deliver consistently at 100% speed
- A Little Less: 2~5 ft further than the max operable range
  - Packets deliver slightly slower → maybe like 50% slower
  - Any more than this and messages practically crawl to the leader node
- Theoretically, range could be doubled if a third node was placed at the same distance from the second node as the latter is from the leader
  - Additionally, painlessMesh supports relayed messages so tests involving LoS blockage for the leader but not the second node should be made.





Next Steps

- Testing 3 Node
  - Considering new max range with third node
  - Considering relay around LoS blockage
- Testing with signal jammers
- What to do with data from follower (+leader) cameras?
  - Point cloud to show accuracy? (Albeit not quite the point of the project but shows reliability of the system)